

Sections and Chapters

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Part I

History of LuaTeX

0.1 What is it and what makes it so different?

LuaTeX is a *toolkit* it contains sophisticated software tools and components with which you can construct (typeset) a wide range of documents. The subtitle of this article also poses two questions about LuaTeX: What is it and what makes it so different? The answer to “What is it?” may seem obvious: “It’s a TeX typesetting engine!” Indeed it is, but a broader view, and one to which this author subscribes, is that LuaTeX is an extremely versatile TeX-based document construction and engineering system.

0.1.1 Explaining LuaTeX: Where to start?

The goal of this first article on LuaTeX is to offer a context for understanding what this TeX engine provides and why/how its design enables users to build/design/create a wide range of solutions to complex typesetting and design problems perhaps also offering some degree of “future proofing”

Chapter 1

LuaT_EX: Background and history

1.1 Introduction

LuaT_EX is, in T_EX terms, “the new kid on the block” despite having been in active development for over 10 years.

1.1.1 LuaT_EX: Opening up T_EX’s “black box”

Knuth’s original T_EX program is the common ancestor of all modern T_EX engines in use today and LuaT_EX is, in effect, the latest evolutionary step: derived from the pdfT_EX program but with the addition of some powerful software components which bring a great deal of extra functionality.

How LuaT_EX processes `\directlua`: A first look

The `<code>` provided to `\directlua{<code>}` is first converted to tokens using the processes and calculations discussed above; that sequence of tokens is stored in a token list.